

Shower Device

Happy Campers

Element H: Prototype Testing and Data Collection Plan

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

We will test our prototype. Our prototype should be waterproof, sturdy - holds up to 10lb, elevated off the floor, should be small <6in deep, <12in wide, fits 5 bottles and a razor or an external shower caddy. This will give us information on how well the prototype meets our design requirements.

- a. To test waterproof:
 1. Direct a stream of water at the prototype.
 2. Record if the prototype collects the water within its pores.
 3. Repeat steps 1-2, four additional times.
- b. To to test sturdiness:
 1. Place mass of 10lb on the prototype
 2. Record if the prototype is still in previous position and still intact

- c. To test elevated off floor:
 - 1. Install prototype.
 - 2. Record if the prototype is elevated off the floor.
 - 3. Repeat steps 1-2, four additional times.
- d. To test dimensions
 - 1. Measure the depth and measure the width.
 - 2. Record the dimensions.
 - 3. Repeat steps 1-2, four additional times.
- e. To test capacity:
 - 1. Place 5 bottles and a razor on the prototype.
 - 2. Record if the bottles and razor fit on the prototype.
 - 3. Place a shower caddy on the prototype.
 - 4. Record if the shower caddy fits on the prototype.
 - 5. Repeat steps 1-2, four additional times.

Our testing plan will test 60% of our design requirements, at least 3 high priority, and give us the information needed. These tests will inform us on whether the prototype is waterproof, sturdy, elevated off the floor, small enough for full function within the shower, and fits the necessary amount of shower products; 5 bottles and a razor.

Our expert Ms. Z reviewed that our testing plan is in the right format, has accurate data recording, and covers all high priority design requirements. Specifically, Ms. Z pointed out that we should repeat each part of the test procedure multiple times. We incorporated this into our testing plan that we carried out. It helped create a stronger conclusion because the multiple repetitions confirmed the test results.